

DD2380 Artificial Intelligence

Course Practicalities

Intended learning outcomes

After passing the course, the student should be able to

- apply different principles of Artificial Intelligence (AI)
- choose appropriate tools and implement efficient solutions to problems in AI
- integrate tools to design computer programs that show different properties that are expected by an intelligent system
- present, analyse, and entitle an own solution to an AI problem
- reflect on and discuss current social and ethical aspects of AI

in order to be able to

- draw use of methods of artificial intelligence in analysis, design and implementation of computer programs
- contribute to design of an intelligent system in both academic and industrial applications.

Lectures

- **Introduction and Practicalities**
- **AI and society**
- **Problem solving**
  - Search
  - Games
- **Logic and planning**
- **Taming uncertainty**
  - Probabilistic reasoning, Bayesian Networks
  - Hidden Markov Models
- **Decision making under uncertainty**
  - Decision making using MDPs and POMDPs
  - Reinforcement learning
- **Multi agent systems + robotics**
- **Machine learning**
- **Natural language processing and conversational systems**
- **Computer vision**
- **Generative AI and wrap-up**

Examination overview

- RAP1
  - Essay
- TENQ
  - 3 classroom quizzes
  - 6 canvas quizzes
- LAB2
  - Prerequisite: Code of conduct quiz
  - 3 programming assignments (code and learning journal)
    - Search
    - HMMs
    - RL

Essay

Aim and idea: Impossible to talk about AI without talking about societal and ethical issues

- An individual reflection, a group discussion, and a group summary (essay) on risks and benefits of AI
- Will be up today Thursday, October 29th after the lecture and is due on Mon November 9th, 19:00
- Pass/fail

On-campus quizzes 1-3

Aim and idea: Recall and apply the basic concepts and principles; a way to stimulate continuous learning

- There will be a on-campus quiz for each of the 3 main topics
  - enroll in Canvas for one of the slots that will open
- 8 multiple-choice questions
- 45 minutes
- Pass/fail (50% correct answers to pass)
- If you fail
  - During the regular period you can repeat **ONE**
    - New slots will open possibly on 4th and 5th of January (an announcement for enrolling will come a week before)
  - In the re-exam period, you will be able to repeat **each once**

Canvas quizzes Q1-Q6

Aim and idea: Recall and apply the basic concepts and principles; a way to stimulate continuous learning

- Quizzes released online after lectures on a topic
- A quiz must be completed (typically 4-5/6 points)
  - Individually, within 60 min, and with 5 attempts
  - Before the deadline
- Multiple-choice/multiple-answer questions
- Pass/fail

Canvas quizzes: a tip from previous generations

- Do them after each lecture, not all in the end

Programming assignments

- Aim and idea: Get you to work hands-on with the course material

- Each assignment structured into E-D, C, B-A levels
  - Search
  - HMMs
  - RL
- You need to be comfy with Python (prerequisite)

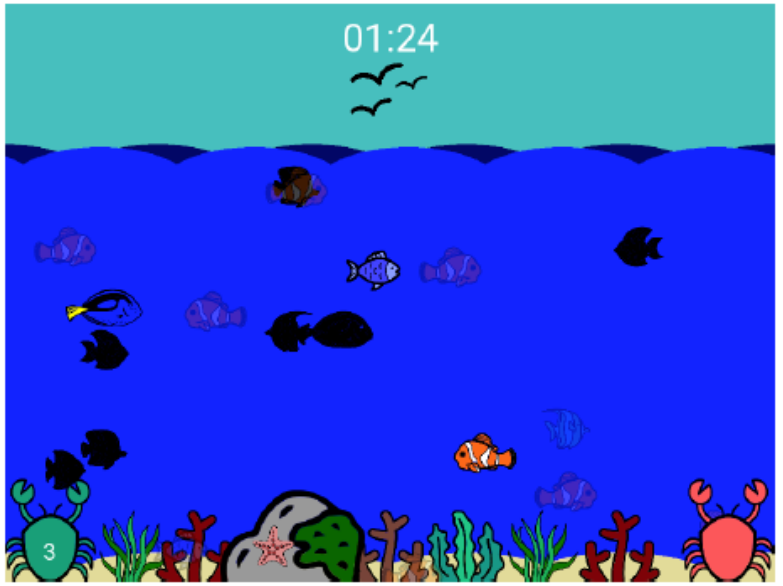


Figure 4.1: Fishing derby

Programming assignments

- Strongly encouraged to be carried out in pairs
- Each assignment is assessed by
  - **Kattis**: checks whether your implementation passes the required tests
  - **Assignment Learning Journal**: explains your solution, design choices, limitations, and reflection questions
- Kattis does **not** directly give the grade
  - It gives the highest-grade range you are eligible for
  - The journal determines the grade within that range
  - https://kth.kattis.com/courses/DD2380/ht26
- Individual oral examination
  - One of Search, HMM, or RL is selected independently for each student
  - You must be able to explain the selected assignment, your code, and your Assignment Learning Journal
  - The oral examination sets your individual maximum LAB2 grade
- Deadline rule
  - Grade A requires all code and Assignment Learning Journals on time
  - If any assignment code or Assignment Learning Journal is late, maximum final LAB2 grade is B

Prog. assignments: tips from previous generations

- Start right away when they are up

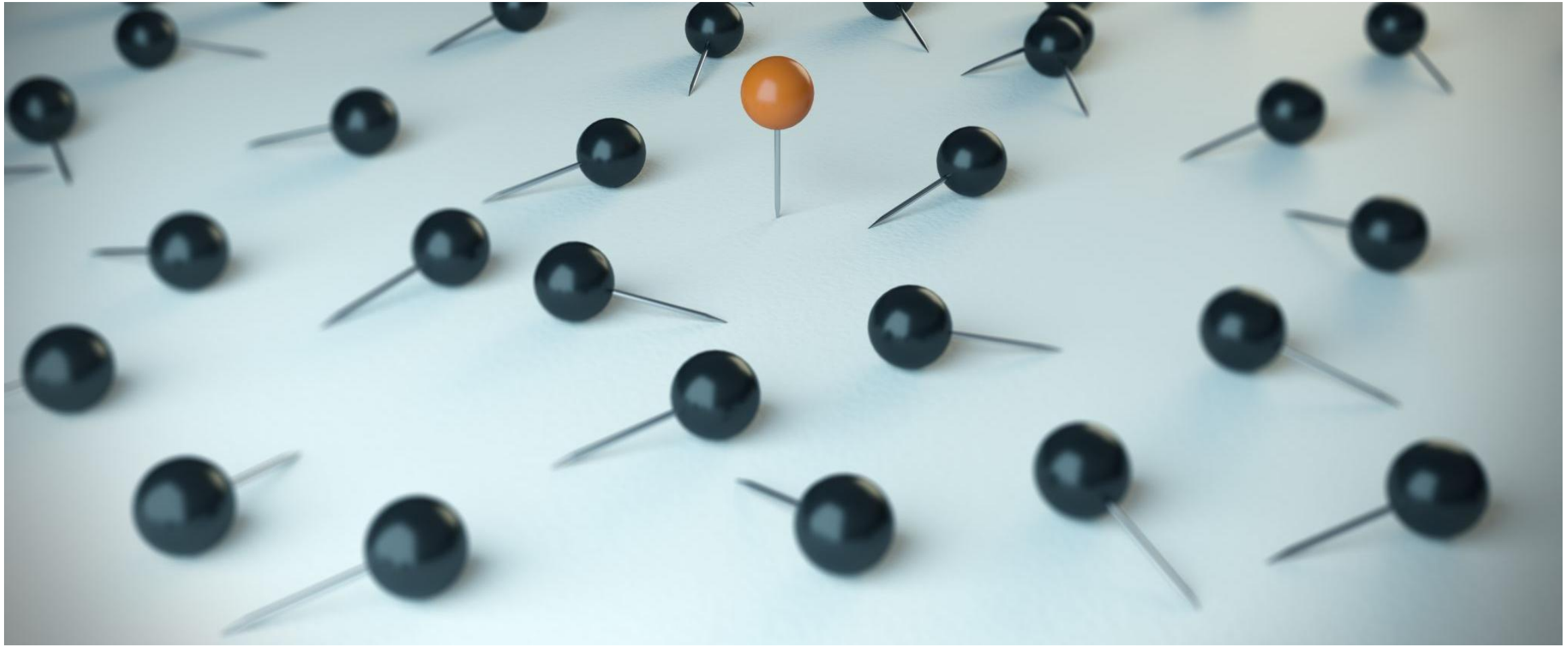
Grading

Assignment task	Evaluation
Essay	Pass/Fail
Quizzes	Pass/Fail
Search	A-F
HMM	A-F
RL	A-F

Pass the course:

- Pass the essay
- Pass all the quizzes
- Get at least E on Search, HMM and RL
- Final grade = LAB2 grade
  - Rounded-down average of Search, HMM, and RL
  - Subject to:
    - Individual oral-examination cap
    - Deadline cap

Practicalities



The course Canvas site

- https://canvas.kth.se/courses/65313
- Events start quarter past.

Crew



André Pereira
(course responsible)
- Problem Solving
- Taming Uncertainty



Iolanda Leite
- Ethics



Petter Ögren
- Reinforcement Learning
- Multi Agent Systems + Robotics



Noemie Jaquier
- Logic & Planning

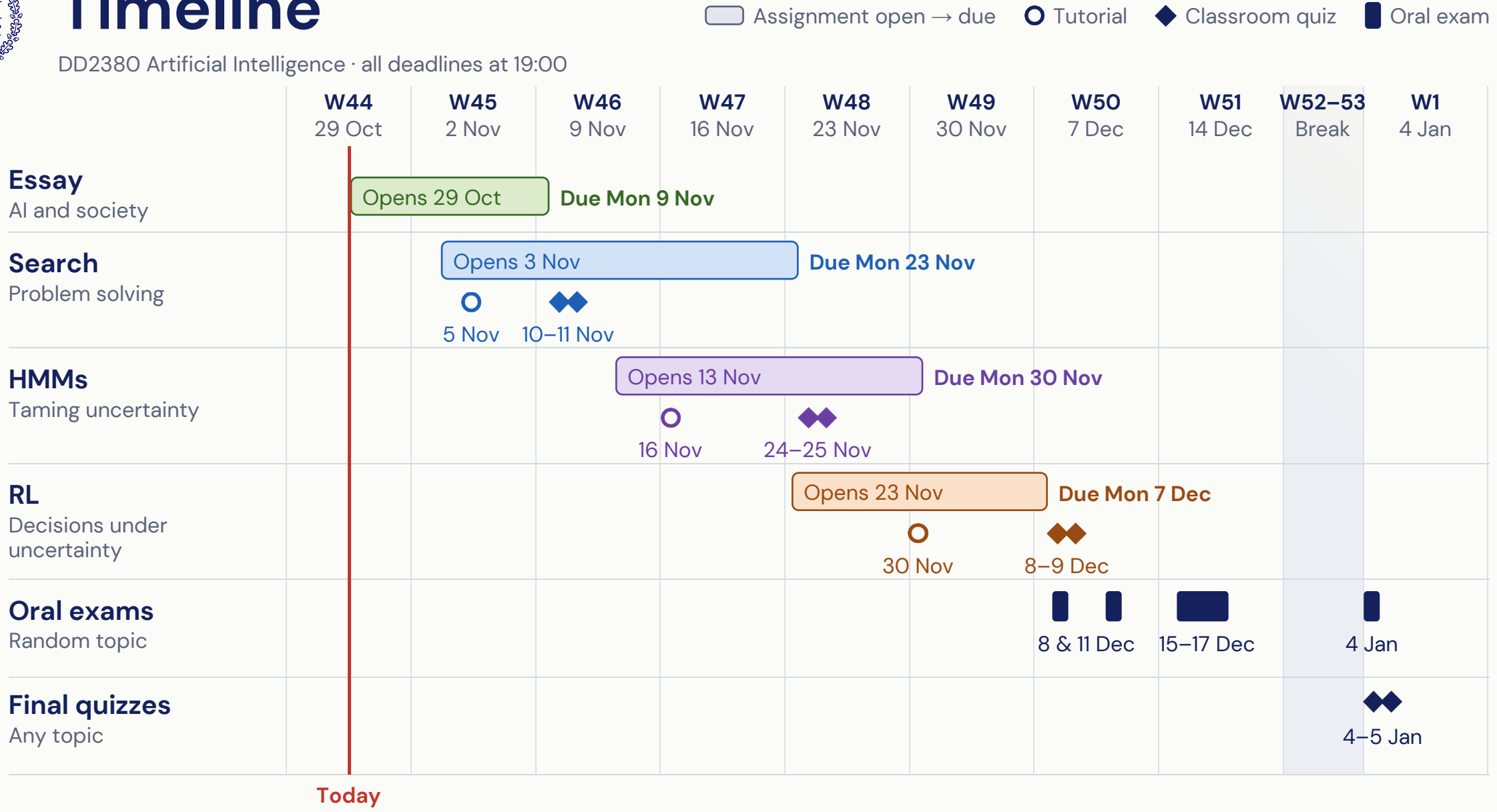
- Guest lecturers
  - Josephine Sullivan, Johan Boye, Gabriel Skantze, Mårten Björkman and Gustav Eje Henter
- 10 teaching assistants (TAs)

Understanding the schedule

- **Lectures:** 14 lectures, each given once; some fundaments and some guest lectures
- **Tutorials:** 3 tutorials, given in multiple timeslots
- **On-campus quizzes:** 3 quizzes, given in multiple timeslots
- **Individual Oral Examinations:** in person
- TAs will have
  - **1-1 consultations ("office hours"):** By using the KTH queue you can have a short Zoom meeting with a TA, we will offer these once per week when the assignments start (mostly Tuesday's)
- **For tutorials, on-campus quizzes, individual oral examinations, you need to sign up -> Canvas, Calendar, Appointments.**

Timeline

DD2380 Artificial Intelligence · all deadlines at 19:00



Where is the course material?

- Lecture slides will appear on Canvas in Modules (we will upload them before the lectures)

- Recommended reading is Russell & Norvig's "Artificial Intelligence A Modern Approach", Pearson New International Edition, 2013.

Communicating with us

- English only

- Important and urgent matters will be posted in **Announcements**
- If we contact you directly, it will be through Canvas Conversations (Inbox) or KTH email

- Primary communication via discussion forums at Canvas
- If you need to contact us via email:

rpl-ai-teachers@eecs.kth.se

Email communication

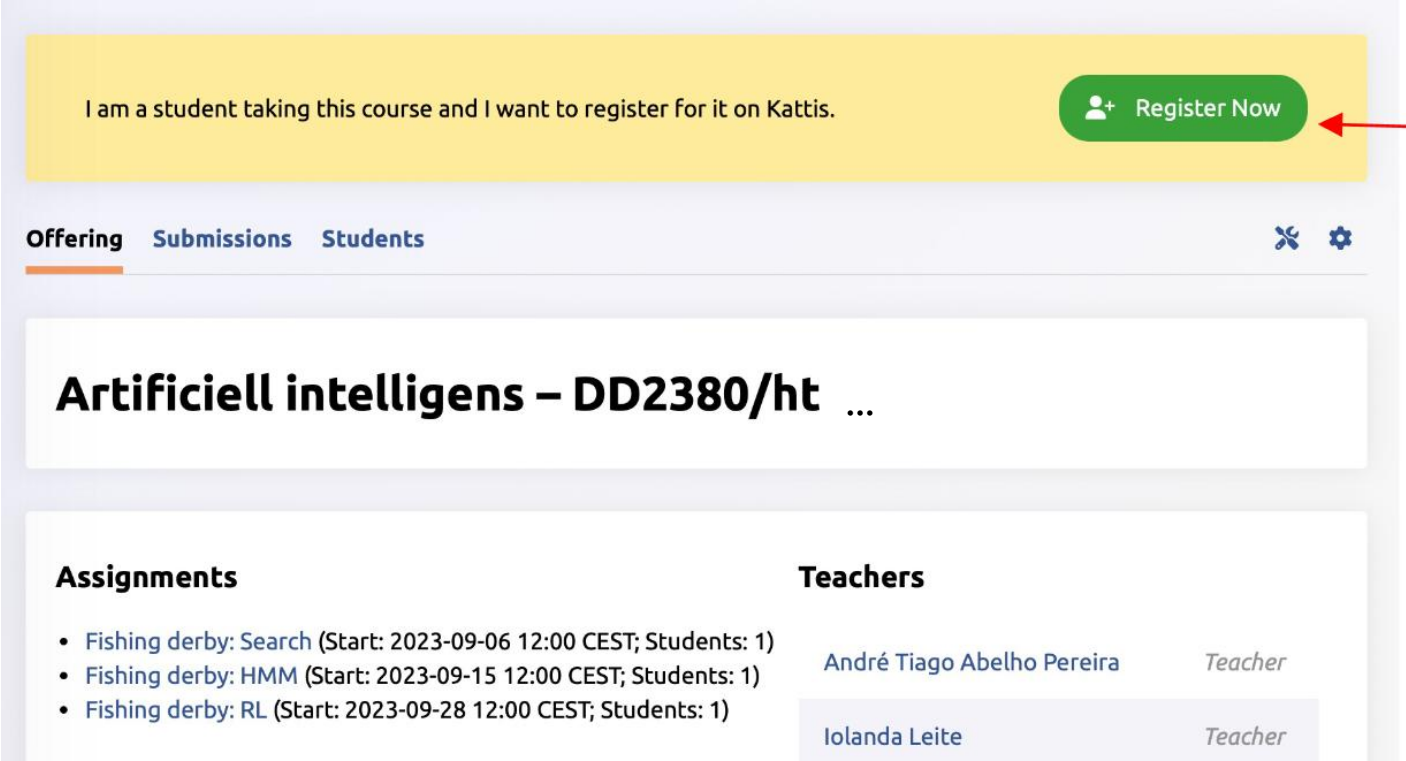
- When to send an email to rpl-ai-teachers@eecs.kth.se:
  - I did a part of the course in HT22 and would like to know what to do to finish.
  - We created an essay group of 3 people and a 4th person just clicked themselves in in the Canvas. The person is not responding, not doing their job. Can you remove them from our group?
  - …
- When not to send an email:
  - When is the next tutorial? How do I sign up for it?
  - I keep getting timeout on HMM assignment. Why?
  - We are three friends and would like to do the programming assignments together. Is it ok?
  - …

Registration for the course

- Make sure to have registered for the semester. Once you get registered, you get to see the course in Canvas, too (within 3 hours or so)

Registration in Kattis

- You need to have an **account in Kattis** (https://kth.kattis.com) and you need to **register for the course in Kattis** to submit assignments
  - Kattis account
  - https://kth.kattis.com/courses/DD2380/ht26 (make sure to login first)



Signing up for tutorials and labs

- Canvas -> calendar -> appointments
- You may need to click on “view appointments” or similar.

TODO Now

- Register for the course (web)
- Go to Canvas!
  - Go to Account -> Notifications, consider switching on Announcement, Conversation Message, Appointment Availability as right away, or daily summary
  - Go to Courses -> DD2380 HT26 (AIHT26_2), checkout the Home page, the Syllabus, the Grading criteria
  - Do your Code of conduct quiz now!
- Get access to Kattis
- Make sure you know your Python

Already some real stuff this week and the next

- After today's lecture on Ethics
  - Start on your essay (due on Nov 9th at 19:00)
- After next Tuesday's lecture on Adversarial Search
  - Start on your first programming assignment (due on Nov 23rd at 19:00)

Questions?